Step 9 · Production Requirements

Assumed: 1,000 active users (~400 a day), ~100 sessions at peak · 4,000 asks a day, 3–6 in flight (designed for 20) · 12,000 searches a day (designed for 10/s) · 2M part revisions, 2–5k releases a day · 2026-10-28

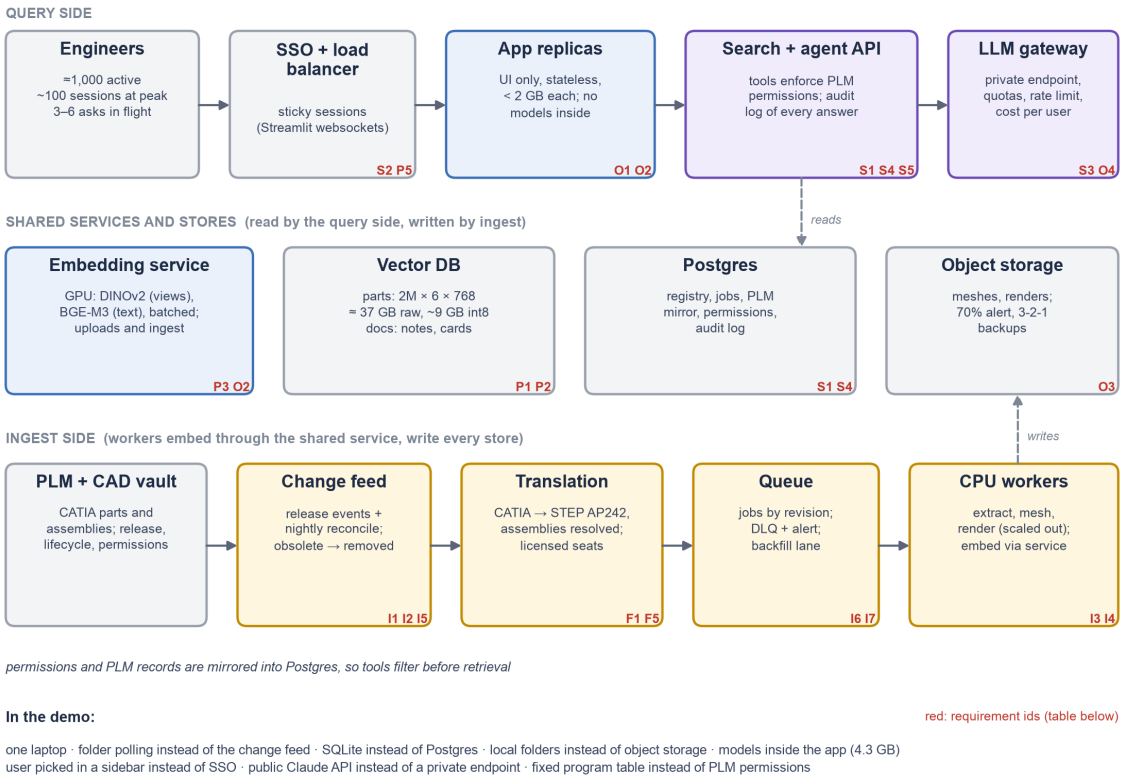


Figure 1. The system at production scale; red ids are the requirements below.

Requirements

Id	The system shall … (target)	Demo	Production
F1	find shape-similar parts for a part number or an uploaded CAD file	yes (STEP)	CATIA via the PLM translation service
F2	search records and notes in Korean and English	yes	same design, enterprise vector DB
F3	cite every fact in an answer (100%)	yes	links into PLM
F4	tell revisions apart: open vs fixed later	yes	revision model from PLM
F5	handle assemblies (CATProduct, thousands of children)	no	resolve references, index structure
F6	aggregate across many records (fix history, counts)	yes	structured reason codes
P1	text search p95 < 1 s at 2M parts	partly: 0.48 s at 1.6k	load test at full size
P2	shape search p95 < 1 s at 2M parts	partly: 0.04 s at 1.6k	int8 / mean vector, MaxSim re-rank
P3	upload → similar parts p95 < 10 s	partly: 1.1 s, max 5.6	GPU service; assemblies async
P4	Ask: progress < 2 s, answer p95 < 30 s	partly: 7.5 s median	stream the answer
P5	20 asks in flight, 10 searches/s, no degradation	no	load test, replicas, rate limits
I1	a release is searchable < 1 h (aim 15 min)	partly: 16 s, folder polling	PLM change feed
I2	no release missed (0 a month)	partly	nightly reconciliation with PLM
I3	ingest 5k revisions a day with margin	partly: ~95k/day on 1 worker	CPU + GPU workers
I4	backfill 2M parts < 2 weeks	no: ~21 days on 1 worker	~10 workers + translator seats

Id	The system shall … (target)	Demo	Production
I5	obsolete or deleted parts leave the index < 24 h	no	lifecycle events
I6	conversion ≥ 98%, every failure logged with an owner	yes: 98.0%, DLQ	alerts, owners
I7	a model change re-embeds without downtime	yes: versioned backfill	side-by-side index, switch
Q1	search usefulness measured on part families	partly: P@5 0.35, no families	PLM families, clicks
Q2	Q&A ≥ 95% on a gold set, 0 invented ids	yes: 97–98%, 0	gold set from real questions
Q3	every change regression-tested	partly: by hand	automatic, with a cost cap
Q4	honest when it can't answer (0 blank answers)	yes	keep in the gold set
S1	PLM permissions enforced before retrieval (0 leaks)	yes: 0 in 204, programs only	real permissions, synced
S2	users log in with company identity	no: sidebar	SSO, groups from the IdP
S3	CAD/PLM data stays within approved bounds	partly: public LLM API	private endpoint or on-prem
S4	every question and answer auditable	partly: traces not kept	audit log, retention
S5	text in records or files can't steer the agent	partly	red team; tools stay read-only
O1	99.5% available in business hours	no: one laptop	replicas, health checks
O2	app process < 2 GB	no: 4.3 GB	models as a GPU service
O3	storage alert at 70%, 3-2-1 backups	partly: 1.49× STEP measured	drop .stl (→ 0.83×)
O4	LLM cost known and bounded	partly: $0.023/question	quotas, dashboard (~$2k/month)
O5	latency, errors, ingest lag, cost observable	partly: offline	metrics, tracing, alerts

Risks the numbers predict

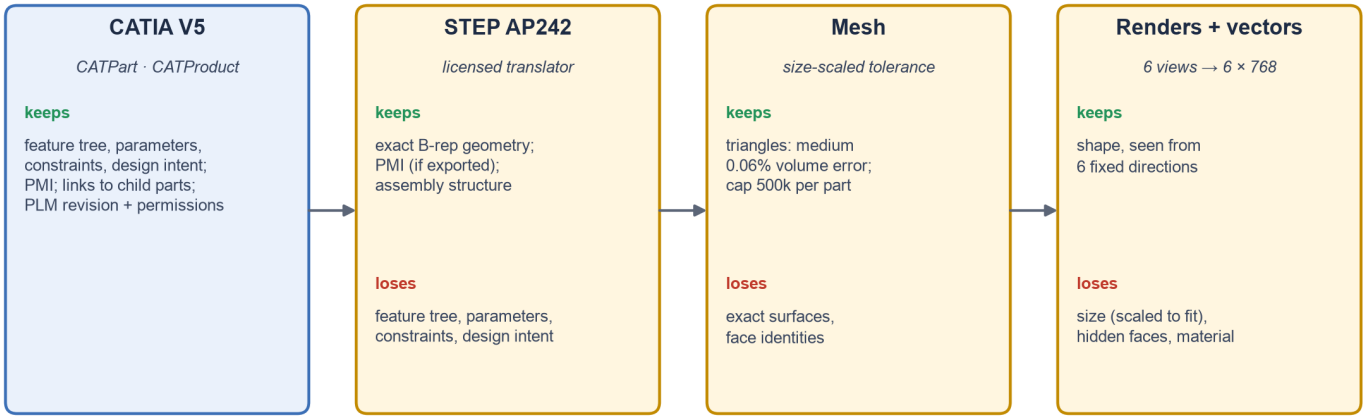
Topic	What the numbers say	Design
Load balancing	Streamlit keeps each session on one websocket; a replica with the models in it is 4.3 GB	sticky sessions; app replicas without models; one shared GPU service
LLM rate limits	a question: ~6.5k input tokens (90% cached), 570 output, 2 requests; hard ones 3×. At 1 ask/s: ~120 requests, ~400k input and ~35k output tokens a minute	check the endpoint's limits and whether cached reads count; queue with backoff; per-user quota
Cost	$0.023 a question → ~$90 a day, ~$2k a month at 4,000 asks a day	quota per user; alert on spikes; prompt cache kept warm
Sharding	2M × 6 × 768 floats ≈ 37 GB; int8 ≈ 9 GB; mean vector only ≈ 6 GB	one node fits: replicate for availability, shard only near 10× the corpus
Caching	the prompt cache already serves ~90% of input tokens	cache searches and answers per permission set, never across users (a shared cache leaks); upload embeddings by sha256
Misuse	a user may pull a whole program through many questions; notes and supplier files can carry injected text	quotas and anomaly alerts on the audit log; read-only tools; ids checked

Unknowns to find out, not guess

Unknown	How to find out
the real query mix and the shape of the peak	pilot with 30–50 engineers; logs
search quality on real part families	label families from PLM; click feedback
translation speed and failures on CATIA assemblies	run the translator on one program's parts
latency and limits on the approved LLM endpoint	load test with recorded questions
the permission model (export control, supplier-specific)	workshop with PLM admins
how it can be attacked	red team before the pilot

CATIA at scale

My understanding is that vehicle design runs mainly on CATIA V5 (anonymous mentor answers, 2024; to confirm). The demo starts from STEP; this is what changes when the source is CATIA in a PLM vault.



My bracket (FreeCAD, parametric): exported to STEP AP242 and read back, the feature tree and its parameters were gone, only the geometry was left, and it was exact: volume 13,967.3 mm³, as calculated by hand. So the pipeline keeps PLM metadata next to the STEP: design intent is not in the file the AI reads.

Figure 2. What each format keeps and loses on the way to a search vector.

CATIA fact	What the demo showed	What the pipeline must do
A CATProduct references many CATParts by link; a child can be missing	ABC files hold several solids; 65% were copies of a solid in the same file	resolve references; index each unique CATPart once with its instance count; a missing child is logged and the assembly kept as partial, not dead (F5)
Batch translation needs licensed translators; seats limit throughput	the pipeline itself: 0.91 s per part, 1 worker	size translator seats for the 2M-part backfill; translation is the likely bottleneck: measure it on one program first (I4)
Body panels are class-A surfaces, not solids	9 of the 10 dead files were 'surfaces only'	accept surface bodies: skip the volume gate, render the surfaces, mark them as surface parts
Parts range from fasteners to body sides	0.1 mm – 9.7 m; an M3 thread reached 331k triangles	keep the size-scaled tolerance and the triangle cap; separate detail levels for viewing and ML
STEP comes as AP214 or AP242	ABC files: AP214 (501); my bracket: AP242; schema logged per file	export AP242 to keep PMI (dimensions, tolerances) and assembly structure
Design intent lives in the feature tree	my bracket: tree gone in STEP, geometry exact	keep PLM metadata beside the geometry; never infer intent from STEP
Revisions live in PLM, not in files	identity = file + sha256	identity = item id + revision; sha256 only to skip unchanged work; ingest released revisions
CGR / 3DXML / JT are lightweight tessellations	coarse .glb for the 3D viewer	fine for viewing; similarity needs exact geometry from STEP, tessellated the same way for every part